

CabsOnline Part 2

Web Development Assignment

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1. Public URL

Production URL:

<https://webdev.aut.ac.nz/~vpk8065/assign/part2/> - Main page booking form

<https://webdev.aut.ac.nz/~vpk8065/assign/part2/#/admin> - I used a hash router so hash symbol needs to be inserted before the admin page.

Note: This is the primary deployment of the application which connects to the AUT webdev server, getBooking.php is the admin.php in Part 1, same thing with createBooking.php and to booking.php.

2. Technology Stack

Frontend Technologies

- React.js (Version 19.2.6)
- Vite (Version 8.0.13)
- JavaScript (ES6 Modules)
- HTML5
- CSS3

Libraries and Packages

- React DOM (Version 19.2.6)
- React Router DOM (Version 7.15.1) — used for client-side routing and navigation
- React Leaflet (Version 5.0.0) — used for interactive map integration
- Leaflet (Version 1.9.4) — JavaScript mapping library

Development and Build Tools

- ESLint — code linting and formatting
- @vitejs/plugin-react — React integration plugin for Vite
- npm — package manager
- GitHub — version control and repository hosting
- Visual Studio Code — development environment

Map Provider

- OpenStreetMap using Leaflet and React Leaflet

Backend

- PHP – used for backend API and server-side processing
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3. How to Run and Build the Project Locally

Prerequisites

Make sure the following are installed

- Node.js
- Npm

Project Structure:

Part2_vpk8065.zip/

api/

assignBooking.php - Assign booking to drivers

createBooking.php - Create new booking

getBookings.php - Retrieve booking data

trackBooking.php - Track booking status

src/

App.jsx - Root component with routing

main.jsx - React entry point

index.css - Global styles

assets/

Logo.png - Application logo

components/
aboutSection.jsx - About section component
addressSearch.jsx - Address search functionality
adminDashboard.jsx - Admin dashboard panel
bookingForm.jsx - Booking form component
bookingTracker.jsx - Booking tracking component
contactSection.jsx - Contact information section
driverSelector.jsx - Driver selection component

styles/
- CSS and styling files

.gitignore - Git ignored files
eslint.config.js - ESLint configuration
index.html - Main HTML entry point
package-lock.json - Locked npm dependencies
package.json - Project dependencies and scripts
README.md - Project documentation
vite.config.js - Vite build configuration

Setup:

1. Extract the zip and navigate to the project folder:
 - cd Part2_vpk8065
2. Install dependencies:
 - npm install
 - npm install leaflet

Note: This will install the following :

- React
- React DOM
- React Router DOM
- React Leaflet
- Leaflet
- Vite
- ESLint
- Other required development dependencies

3. Start the development server:
 - npm run dev

The app will be available at <http://localhost:5173/>

Building for Production

To build and deploy to the webdev server:

- npm run dev

Upload the generated folder called dist/ to htdocs/assign/part2/, via FileZilla on MacOS or WinSPC for windows. I created a part2 directory inside the assign folder. The vite.config.js base path is set to /~vpk8065/assign/part2 this should be strictly followed so that it resolves correctly on the webdev server.

4. Microservice API Endpoints

All API endpoints are PHP scripts running on the aut webdev server which in production this will only work in AUT webdev server.

Local API Endpoints

<http://localhost:5173/~vpk8065/assign/part2/api/createBooking.php>

<http://localhost:5173/~vpk8065/assign/part2/api/getBookings.php>

<http://localhost:5173/~vpk8065/assign/part2/api/trackBooking.php>

<http://localhost:5173/~vpk8065/assign/part2/api/assignBooking.php>

Note: this local API endpoints does not for production because PHP script runs only AUT server.

Remote API Endpoints

<https://webdev.aut.ac.nz/~vpk8065/assign/part2/api/createBooking.php>

- Handles HTTP POST requests to create new booking records by validating incoming booking data, processing request parameters, and

storing booking information in the backend database. Returns JSON responses indicating booking creation success or failure.

<https://webdev.aut.ac.nz/~vpk8065/assign/part2/api/getBookings.php>

- Processes HTTP GET requests to retrieve booking records from the backend database and returns structured JSON data for frontend rendering and booking management.

<https://webdev.aut.ac.nz/~vpk8065/assign/part2/api/assignBooking.php>

- Implements booking assignment logic by associating available drivers with booking records, updating assignment status in the database, and returning assignment confirmation responses through the API.

<https://webdev.aut.ac.nz/~vpk8065/assign/part2/api/trackBooking.php>

- Implements booking assignment logic by associating available drivers with booking records, updating assignment status in the database, and returning assignment confirmation responses through the API.

5. Feature Descriptions

Auto-Complete Address Search

Implements dynamic address autocompletion using the OpenStreetMap API to provide real-time location suggestions based on user input. The feature improves usability by reducing manual address entry and enhancing location accuracy for booking creation.

Driver Assignment System

Implement backend booking assignment functionality that associates available drivers with active bookings. The system updates assignment status dynamically and supports administrative booking management operations.

Booking Monitoring / Tracking Status

Provides real-time booking monitoring and tracking functionality through API-based status retrieval. Users and administrators can monitor booking progress, assignment updates, and current booking states within the application interface.

Modern Booking Interface

Implements a responsive and user-friendly booking interface using React components and dynamic UI rendering. The interface improves user experience through structure layouts, interactive forms, real-time updates, and intuitive navigation for booking operations.

6. Testing Instructions

- Example booking references:
 - BRN00007
 - BRN00008
 - BRN00009
 - BRN00011
 - BRN00012

☐	  	7	BRN00007	Christian	0220196070	22	22	Grassways Avenue	Pakuranga	Howick	2026-05-08	22:30:00	2026-05-08 22:12:18	unassigned	NULL
☐	  	8	BRN00008	Jeline Piniac	02108433796	20	20	Roosevelt Avenue	Onehunga	Pakuranga	2026-05-26	21:00:00	2026-05-26 20:03:59	assigned	D001
☐	  	9	BRN00009	Celeste Cantos	0277122001	22	22	Grassways Avenue	Pakuranga	East Tamaki	2026-05-26	21:30:00	2026-05-26 20:13:27	assigned	NULL
☐	  	11	BRN00011	Christian Cantos	0220196070			Pakuranga, Auckland, 2010, New Zealand	20, Roosevelt Avenue, Oranga, Onehunga, Maungakie...		2026-05-28	16:17:00	2026-05-28 15:17:42	assigned	D003
☐	  	12	BRN00012	Emmanuel	0211586825			22, Grassways Avenue, Pakuranga, Howick, Auckland...	5/320, Pakuranga Road, Pakuranga Heights, Howick, ...		2026-05-28	20:18:00	2026-05-28 19:18:43	assigned	D002

Note:

The first bookings was used in Part 1 where unnumber, snumber, stname needs to be filled out, but because I implement a modern interface I didn't include inputs for those 3 fields instead I made it not required to be filled because apparently when I tried to use my admin.php codes in my part 2 getBookings.php it somehow crashed and can't access my database because in my admin.php this fields are required to be filled, so in my getBookings.php I takeout this validation from those fields.

- Example Driver IDs
 - D001
 - D002
 - D003
 - D004

Note:

Driver assignment system handles only in the admin dashboard so when a booking requests was created in the admin dashboard this new request can be seen just by clicking the “Search Bookings” and then unassigned bookings will appear and then from that there’s a selection of driver in the right side of the admin dashboard, after driver is selected for it to be set to assigned status “Assign” button must be click. Then, it will update the status to Assigned and driver id will be attached to that booking reference.

Testing Each Page:

Booking Page: Enter your name, phone, and search for pick up location, and destination location, and and autocomplete address will pop-up, just click it. Set your date and time when this booking will occur. Booking Tracking is attached to the booking page so bookers can either click the navigation bar for tracking and it they will be directed to the section of booking tracking, just enter your booking reference (E.g. BRN00001) and you’ll see your booking if it’s already assigned or not.

Admin Page: To search for unassigned bookings just leave the search bar empty and click “Search Bookings” all unassigned bookings will appear and for it to be assigned and assigned a driver, you need to select a driver from the selections of driver beside the search bar and select a driver and then and right side of the booking request information there will be a “Assign” button which you will click for it to be assigned.

7. Any limitations or known issues

- The application currently uses a simplified backend implementation and does not include full authentication or user account management features.
- Real-time booking updates are limited and may require manual refresh depending on API response timing.

- OpenStreetMap address search results may occasionally return incomplete or inaccurate location suggestions depending on the availability of external map data.
- Fare estimation calculations are approximate and may not fully reflect real-world transport pricing conditions.
- Driver assignment functionality currently uses simplified assignment logic and does not implement advanced scheduling or route optimisation algorithms.
- The application relies on internet connectivity for OpenStreetMap tile rendering and API-based location services.
- Backend PHP APIs currently use local development configurations and may require additional security, validation, and deployment optimisation for production environments.
- Error handling and form validation are implemented at a basic level and could be further improved for larger-scale deployment.
- The system has been primarily tested on modern desktop browsers and may have minor responsiveness issues on older mobile devices or unsupported browsers.

8. Reflection on AI-Supported Development Process

Artificial Intelligence tools specifically ChatGPT(OpenAI) and Claude(Anthropic) were used throughout the development process to assist with coding, debugging, UI design improvements, and technical documentation. AI support helped accelerate development by providing explanations for unfamiliar concepts, suggesting solutions for runtime errors, and assisting with React component structure and PHP API integration.

During development, AI tools were particularly useful for:

- Debugging React and Vite configuration issues
- Assisting with API communication between the React frontend and PHP backend
- Improving component structure and reusable UI design
- Providing guidance for OpenStreetMap and React Leaflet integration
- Assisting with responsive interface design and feature implementation

AI assistance significantly improved development efficiency and reduced time spent troubleshooting configuration and syntax-related problems. However, all generated

code and suggestions were manually reviewed, tested, and modified to ensure correctness, maintainability, and compatibility with assignment requirements.

The development process still required independent problem-solving, debugging, testing, and integration work to successfully implement application features and ensure stable functionality across the frontend and backend systems.